① → Availability guarantee = 99%
→ 12 hours per day for 30 days
→ downtime = 11 hours.
Did the Provider voilate the SLA?

Solution

→ Total Service time = 12 x 30 = 360 hours

→ Total uptime = Service time - downtime
→ Total uptime = 360 - 11 = 349

$$\text{Availability Percentage} = \frac{\text{Total uptime}}{\text{Total Service time}} \times 100$$

$$\text{Availability \%} = \frac{349}{360} \times 100.$$

$$\boxed{= 96.94\%}$$

96.94% < 99% yes Provider voilate the SLA.

② Allowed Downtime Calculation

→ guarantees 99.9% for a Service running 24 hours per day.
→ Calculate the maximum downtime allowed in 30 days.

→ Total Service time = 24 x 30 = 720 hours
→ Missing percentage = ~~99.9%~~ 100% - 99.9%
= 0.1%

So the 0.1% of 720 hours is.

max. Downtime allowed = $\frac{0.1\% \times 720}{100}$ = $\boxed{0.72 \text{ hours}}$

OR

$\boxed{0.72 \times 60 = 43.2 \text{ minutes}}$

(3) Availability Percentage Calculation.

→ System runs 24 hours per day
→ for 30 days
→ During this period, the service was unavailable for 5 hours.
→ Calculate actual availability Percentage.

Solution

→ Total service time = 24 × 30 = 720 hours.

→ Service available or uptime = 720 − 5 hours = 715 hours.

actual availability Percentage = $\frac{\text{Uptime}}{\text{Total Service time}} \times 100$

$= \frac{715}{720} \times 100$

actual availability % $\boxed{= 99.30\%}$

(6) Allowed Downtime per Day

→ Providers offer 99.5% availability
→ what is the maximum downtime allowed per day if the service runs 24 hours daily?

Missing percentage = 100% − 99.5% = 0.5%

Downtime allowed per day is equal to missing percentage of total time

So, 0.5% of 24 hours =

$$\frac{0.5}{100} \times 24 = 0.12 \text{ hours}$$

OR

7.2 minutes

∴ 7.2 minutes total downtime allowed per day

(7) Finding Downtime From Availability

→ A System guarantees 99% availability

→ Total time in a month = 720 hours.

→ What is the maximum downtime allowed?

Solution

Remaining percentage = 100% − 99%
= 1%

So, maximum downtime allowed $= \frac{1}{100} \times 720$

$= \boxed{7.2 \text{ hours}}$

Scenario Questions:

A Company signs an SLA with a Cloud Provider guaranteeing 99% availability for a 30-day service period. The Service runs 12 hours per day and costs $50 per day.

During the month, outage occurred:

→ 30 minutes

→ 1 hour

→ 1 hour 30 minutes

→ 45 minutes

Service Credit Policy

Monthly uptime	Credit
<99%	10%
<98%	25%

(B) Tasks

① Calculate total downtime

$0.5 + 1.5 + 0.75 + 2 = \boxed{4.75 \text{ hours}}$

② Calculate actual availability

Total Service time = $30 \times 12 = 360$

$$\text{Actual availability} = \frac{360 - 4.75}{360} \times 100$$

$$= \boxed{98.68\%}$$

③ Check SLA Violation

$98.68\% < 99\%$

yes Provider violate the SLA. now Customer achieve 10% Credit.

④ Determine Service Credit.

Total Service Cost = $50 \times 30 = \$1500$

Service credit amount = $10\% \times 1500 = \$150$